

(Please write your Enrolment No. immediately)

Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

3rd Semester, November, 2022

Paper Code: ES-201

Time: 1½Hrs.

Subject: Computational Methods

Max. Marks: 30

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

Q.No.	Questions	Max. Marks	CO(s)
1(a)✓	Find a root of the equation $x^3 - 4x - 9 = 0$ using Bisection method.	2.5	CO1
1(b)✓	Define the Absolute, Relative and Truncation Errors.	2.5	CO1
1(c)	Evaluate $\Delta^3[(1-6x)(6-x)(5-3x)]$	2.5	CO2
1(d)✓	Use Trapezoidal rule to estimate the integral $\int_0^1 \frac{dx}{1+x}$.	2.5	CO2
2(a)✓	Perform four iteration of the Newton's Raphson method to obtain the approximate value of $(17)^{1/3}$ starting with the initial approximation $x_0=2$.	5.0	CO1
2(b)✓	Minimize $f(x) = x^2 + \frac{54}{x}$ in interval (0,5). Using Fibonacci search method. $n=4$	5.0	CO1
3(a)✓	For the data X :- -2 -1 0 1 2 3 F(x):- 15 5 1 3 11 25 Construct the Newton's forward difference table and find $f(-1.5)$.	5.0	CO2
3(b)✓	Find the value of $f(9)$ from the given table X: 5 7 11 13 17 F(X) 150 392 1492 2366 5202	5.0	CO2
4(a)✓	Round off the numbers 86.5250 and 37.46235 to four significant figures and compute Absolute error, Relative Error and Percentage error in each case.	5.0	CO1